The Wow! signal faded while it was in the beam: a recorder-aware reanalysis, a frequency-reference discrepancy, and constraints on natural and artificial origins

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Abstract

The 1977 "Wow!" signal survives as six quantized signal-to-noise values (6EQUJ5) from one 10 kHz channel of the Ohio State Big Ear telescope. We re-derive what those six numbers constrain with a forward model of the instrument that includes, for the first time, the recorder algorithm (N50CH running baseline and σ, floor quantization, 10 s integration in 12 s records) and an interval-censored likelihood. Three results are new. (i) The profile is asymmetric: the leading wing is 1.7 units brighter and the trailing wing 1.5 units fainter than any symmetric beam allows; a symmetric crossing reproduces data this skewed in 0.37% of Monte-Carlo trials ($\approx 2.9\sigma$), while a profile that fades with an e-folding time of 17^{+4}_{-3} s starting ≈ 7 s after beam centre fits essentially perfectly. The fade explains the blank second beam 172 s later without additional assumptions and places the detection in the first beam to sweep the source with probability 0.89 (Bayes factor 8.4), i.e. at RA $19^{\rm h}27^{\rm m}55^{\rm s}$ rather than the commonly quoted $19^{\rm h}25^{\rm m}02^{\rm s}$. (ii) The second local-oscillator reading at the peak, under the Galactic-standard-of-rest tracking described by the observatory's staff, implies a frequency reference of 120.00 MHz rather than the 120.1 MHz adopted in the 2025 reanalysis, moving the line from 1420.726 to 1420.826 MHz and $V_{\rm LSR}$ from -74 to -95 km s^{-1} (the 2025 paper itself gives both reference values, and its calibrating clouds post-date the relevant LO change); all candidate velocities are forbidden for circular rotation at $l = 11.6^\circ$ and fall in the velocity range of the compact-clump HVC complex GCN. (iii) The 1977 August 15 continuum chart, digitised here, gives the beam on the night: HPBW 36 ± 1.5 s on both horns and a beam separation of 181 ± 4 s rather than the assumed 172.4 s; against that beam the Wow! core is unresolved and the anomaly reduces to a leading excess of $\approx 10\%$ of peak about 30 s before beam centre, which neither horn's measured response reproduces. Recorder absorption of the first record raises the flux lower limit to $\gtrsim 300$ Jy. Simulations exclude every steady source, including scintillating ones; a stochastic repeater sits at 1.0% likelihood after 193.5 h of follow-up. A natural origin requires coherent 21-cm emission from a cold, dense, tens-of-AU knot pumped by a flare whose geometry forces a circumstellar host and a flare that reached Earth within a minute of the pulse; an artificial origin is technologically undemanding but requires a 10^{-5}–10^{-2} timing coincidence. We estimate posterior odds of roughly 50–60% anthropogenic or instrumental, 25–35% natural transient and $\lesssim 10\%$ artificial extraterrestrial, and identify four archival tests costing hours to days that would reorder them.

1 Introduction

The narrowband event recorded by the Ohio State University Big Ear telescope on 1977 August 15 (local date) remains the most-cited candidate radio technosignature (Ehman, 2010; Gray, 2012). Its properties were set in the 1990s (Ehman, 2010; Gray, 1994) and revised only recently: Méndez et al. (2024) proposed a natural origin in a flaring cold hydrogen cloud; Méndez et al. (2025) re-reduced the archival printouts to obtain a peak flux density $\geq 256 \pm 63$ Jy, a corrected frequency of 1420.726 ± 0.005 MHz (a spectral-axis inversion and a 2nd-LO reference of 120.1 MHz), a 27 s shift in right ascension and an upper limit of $1.9'$ on the source size; and Méndez et al. (2026) published the first account of the Ohio SETI program's archive, including >40 000 narrowband events. Follow-up searches (Gray, 1994; Gray & Marvel, 2001; Gray & Ellingsen, 2002; Harp et al., 2020; Breakthrough Listen, 2022) have found nothing, and Kipping & Gray (2022) showed that a stochastic repeater is in 2.4σ tension with the follow-up record.

All of these analyses fit the six printed values with a symmetric Gaussian and treat the printed numbers as linear samples of the beam pattern. Neither assumption has been tested. Here we build a forward model of the full signal path, fit it with a likelihood appropriate to floor-quantized data, and ask what the data actually constrain (Sect. 2–3). We then simulate the principal origin classes against the same constraints (Sect. 4), work through the physics of the one surviving natural mechanism (Sect. 5) and the requirements of an artificial source (Sect. 6), and argue for a probability assessment (Sect. 7).

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2 Data and forward model

The data. N50CH printed, every ≈ 12 s per channel, the integer part of the signal-to-noise ratio of a 10 s mean of 1 s samples, with 0 blank and 10–35 as A–Z (Ehman, 2010). The sequence 6EQUJ5 therefore means SNR in $[6,7)$, $[14,15)$, $[26,27)$, $[30,31)$, $[19,20)$, $[5,6)$ in six consecutive records, with the records before and after printed blank or "1". Everything else known about the signal is inferred.

The instrument. Big Ear was a meridian-transit instrument with HPBW $\approx 8'$ in RA (36–39 s of time at Dec $-27°$) and $40'$ in Dec, two feed horns whose beams were separated by 172.4 s of RA, and a Dicke switch between them at 79 Hz; the recorded quantity is the difference, and before 1977 October 22 its absolute value was printed, so a steady source produced two positive events 172 s apart. The negative (west) horn's beam swept a source first; the positive (east) horn's beam second (Méndez et al., 2025). The running baseline B (weight 1/6) and σ (weight 1/60) were frozen while a signal exceeded a threshold, which Ehman states was the case for Wow!. We adopt the per-record freezing he describes and test the alternatives.

Forward model and likelihood. A source of brightness $A\,b(t)$ crosses a Gaussian beam of HPBW w centred at t_0; record j averages ten 1 s samples of unit noise starting at $12j$; the printed value is $\lfloor \bar{S}_j \rfloor$. The likelihood of a printed k is $P(k \le X < k+1)$ with $X \sim \mathcal{N}(m_j, 0.316^2)$, where m_j is the model record mean and $0.316 = 1/\sqrt{10}$ is the noise on a 10-sample mean; records before and after are constrained to $X < 2$. Posteriors are computed on grids; model comparison uses maximum log-likelihood, Bayes factors with explicit priors, and Monte-Carlo goodness-of-fit (3000 symmetric crossings through noise, recorder and quantization).

3 What the six numbers constrain

3.1 The profile is asymmetric

The symmetric Gaussian beam reproduces Ehman's unadjusted WOWFIT exactly (SSE 7.53 vs his 7.525), which validates the geometry, but it is a poor fit: maximum $\ln\mathscr{L} = -23.6$ against a best achievable ≈ -0.7, because it predicts 4.8 for the "6" record and 7.0 for the "5" (Fig. 1). Table 1 compares profile models. Allowing different half-widths on the two sides gives $\ln\mathscr{L} = -2.7$ with HPBW 45.3 s rising and 30.7 s falling; a physical pulse that is flat and then decays exponentially from $t_{\rm dec}$ gives -4.7 with $t_{\rm dec} = 46.7 \pm 1.2$ s (≈ 7 s after beam centre), e-folding time 17^{+4}_{-3} s and a beam-side half-width of 44 ± 2 s. The brightness 25 s before beam centre was $2.6^{+0.2}_{-0.6}$ times that 25 s after.

The significance is assessed by simulation. A true symmetric crossing with $A = 31.2$, $w = 38.1$ s, 1 s Gaussian noise, the per-record N50CH filter (threshold 6σ) and floor quantization yields a best symmetric fit as poor as -23.6 in 11 of 3000 trials ($p = 0.0037$). Ehman's SSE of 0.32 was reached only after a post-hoc search over 3^6 adjustments of ± 1 to each point.

Table 1: Profile models fitted to 6EQUJ5 with the interval-censored likelihood.

Model	max $\ln\mathscr{L}$	Fitted parameters
Symmetric Gaussian beam	-23.6	t_0=36.7 s, A=31.2, w=38.1 s
Symmetric $\sin^2 x/x^2$ beam	-33.8	w=39.6 s
Sym. beam, first record absorbed	-10.2	A=35.0, w=38.7 s
Sym. beam, switch-off at 65 s	-9.8	A=30.7, w=40.3 s
Flat pulse, exponential fade	-4.7	$t_{\rm dec}$=46.7 s, τ=17 s, w=44 s
Skewed Gaussian, two half-widths	-2.7	45.3 s rising, 30.7 s falling

Instrumental alternatives. Had the baseline updated every second with weight 1/6, the 6 s time constant would have swallowed most of a 38 s pulse: no amplitude or width reproduces the data ($\ln\mathscr{L} < -4000$ for thresholds of 5–8σ). The data therefore require per-record freezing, as recollected. The only surviving recorder effect is absorption of the first record into the baseline if the freeze threshold exceeded 6σ; it reaches $\ln\mathscr{L} = -10.2$, still 7.5 units short of the skewed profile, and raises the true peak from 31.2 to 35.0 SNR units (Sect. 3.2). An asymmetric beam is the remaining instrumental option: the horns sat $34'$ and $73'$ off the paraboloid axis, so coma produces a leading or trailing shoulder that should differ between horns. Ehman's correlation of 6EQUJ5 against 1994 calibrator transits (0.9905 vs 0.9919 for the two horns) has no power against a 1.5:1 asymmetry. The decisive measurement is the leading and trailing half-width of each horn's beam on the 1977 calibrator transits (OC-230.4, OC-266, OP-282, OF-241, PKS 1628−268) that Méndez et al. (2025) have already digitized.

Consequence: a fading source and the horn. If the asymmetry is intrinsic, the source lost a factor $\sim$2.5 in brightness during the crossing. A pure exponential fade cannot do this—multiplied by a Gaussian beam it merely shifts a symmetric profile—so the decline steepened, as for a pulse ending. That by itself explains why the other beam saw nothing 172 s later, and it bears on which beam detected the signal. Under the flat-then-fading pulse model with a log-uniform prior on pulse duration (1 s to 1 d) and identical priors otherwise, the evidence for detection in the first beam (other beam later) over the second beam (other beam earlier, which requires the pulse to have both begun and ended inside a ≈ 250 s window) is a Bayes factor of 8.4, $P(\text{first}) = 0.89$. The favoured position is therefore the negative-horn one: RA $19^{\rm h}27^{\rm m}55^{\rm s} \pm 3^{\rm s}$, Dec $-26°57' \pm 20'$ (J2000), $l = 11.87°$, $b = -19.42°$. Within the model, fading during the crossing is preferred to no fading by 2.5×10^7; that preference, however, assumes a beam of 38–44 s and is revisited in Sect. 3.3 with the beam measured on the night.

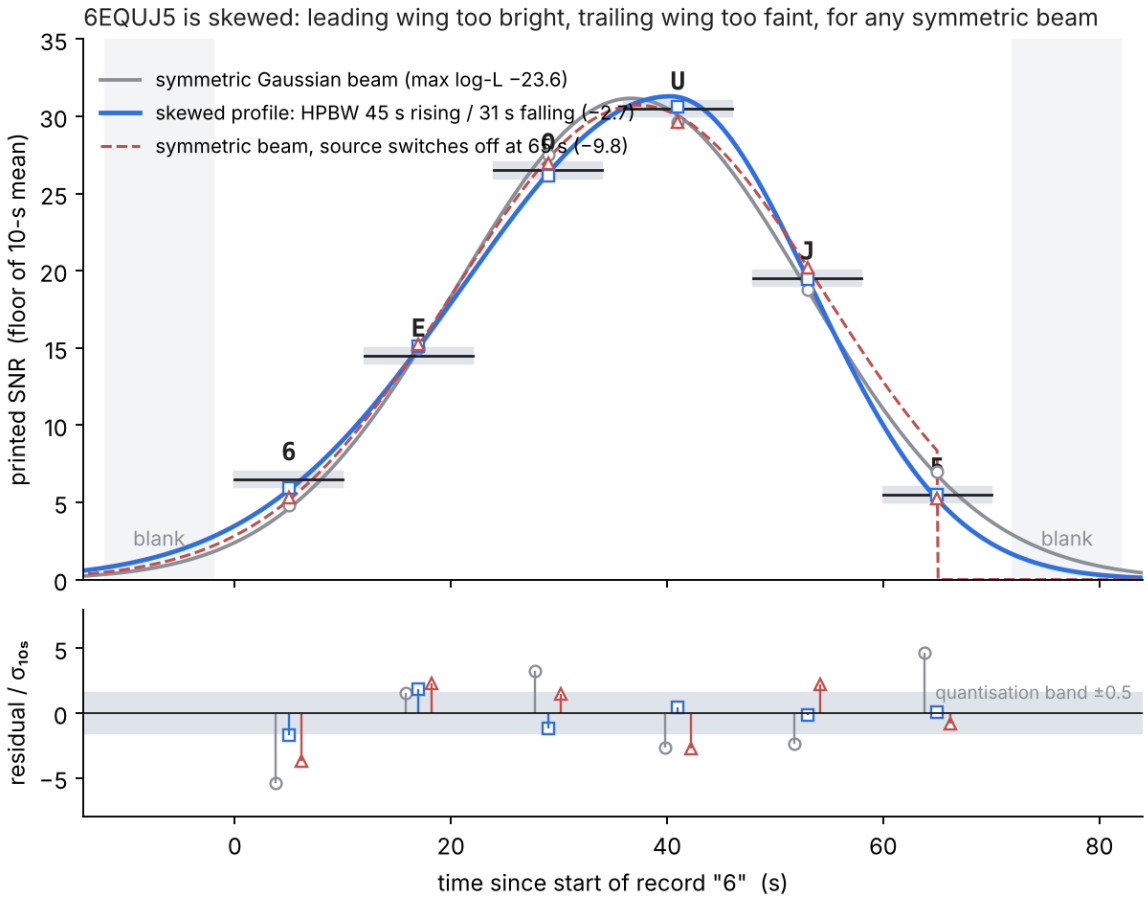


Figure 1: The six printed values as quantization intervals (grey bands), with the best symmetric Gaussian beam crossing (grey), the best skewed profile (blue) and a symmetric beam with the source switching off at 65 s (red). The lower panel shows residuals in units of the per-record noise; the band is the ± 0.5 quantization width. The symmetric model misses the first and last records by -5σ and $+5\sigma$.

3.2 Flux, width and angular size

Within the symmetric model the peak is 31.2 ± 0.3 and the HPBW 37.9 ± 0.5 s; across recorder assumptions the peak is 31–35, so the Méndez et al. (2025) lower limit of 256 Jy becomes 265–300 Jy before any correction for the unknown declination offset (up to $\times 2$). From first principles, $T_{\rm sys} \approx 100$ K and 0.1 K Jy^{-1} give an SEFD of ≈ 1000 Jy and a 10 kHz$\times$10 s Dicke-switched record noise of 4.5–9 Jy, bracketing the 8.5 Jy adopted there.

The angular-size inference is not robust. With the nominal $8.0'$ beam (35.9 s) the quantization-aware posterior on the width implies a resolved source of $2.7' \pm 0.8'$ (quadrature difference); with the $8.6'$ beam (38.6 s) that Ehman measured on OY 372 in 1994 the source is unresolved ($< 0.5'$). The published $\leq 1.9'$ assumes the former beam and a point-like fit. The 1977 beam width measured in Sect. 3.3, 36 ± 1.5 s, implies $2.7'$ ($+0.8'/-1.5'$) if the whole profile is one source, but the core records alone are fitted by a point source once the leading excess is a separate component.

3.3 The beam on the night of the signal

The 8 MHz continuum chart of 1977 August 15 (Dec strip $-26°00'$ set, $-27°$ beam; 3 s time constant; Fig. 2) records the Ohio source OV-221 crossing both horns 50 minutes before Wow!. Digitising the trace with the grid removed, and calibrating the time axis on the 1910/1920/1930 LST marks (2.11 s per pixel), gives three numbers nobody had: a half-power width of 35.7 s for the negative horn and 36.8 s for the positive horn after deconvolving the time constant, i.e. the nominal $8.0'$ beam at Dec $-27°$ (36 ± 1.5 s) rather than the $8.6'$ of the 1994 OY 372 scans; a separation of the two beams of 181 ± 4 s, 5% larger than the 172.4 s derived from the horn spacing and used in every published position; and the per-horn response at the six Wow! record offsets, 0.12/0.36/0.78/0.92/0.67/0.16 (negative) and 0.15/0.64/0.91/0.88/0.48/0.10 (positive, after a ramp baseline across a 0.3-peak baseline shift) against Wow!'s 0.21/0.46/0.85/0.98/0.62/0.18.

Three consequences follow. With the beam fixed at 35–36.5 s, the Wow! anomaly is a *leading excess* rather than a two-sided skew: the records E, Q, U, J are fitted by the measured

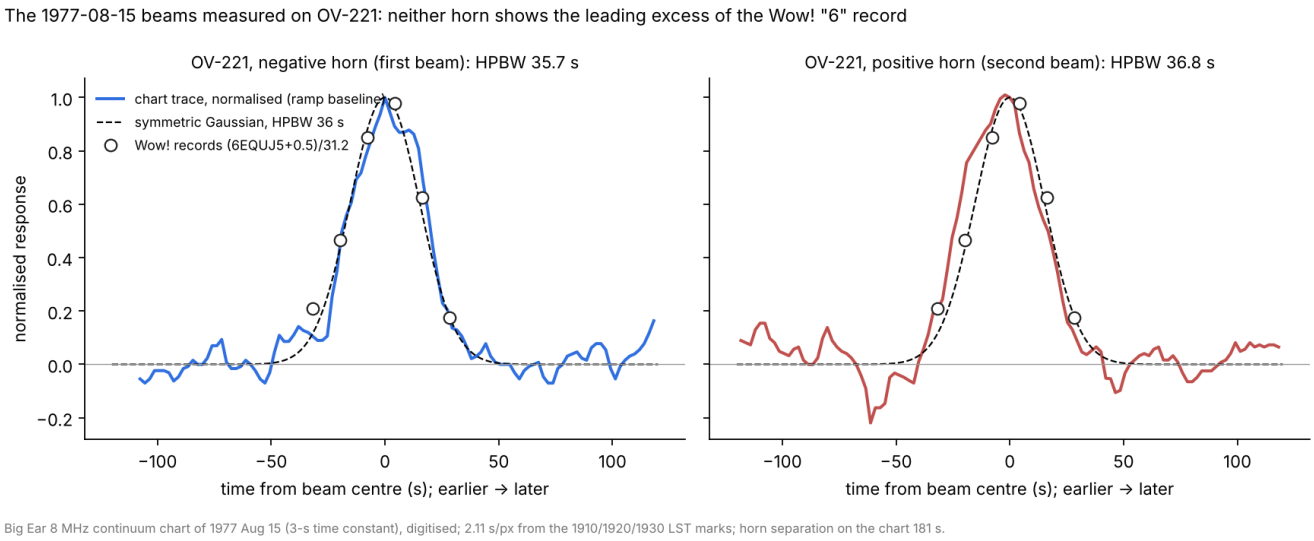


Figure 2: The two beams measured on OV-221 on 1977 August 15 (continuum chart, digitised and normalised; ramp baseline) compared with a 36-s Gaussian and with the six Wow! records at their offsets from beam centre. Neither horn reproduces the Wow! leading record.

beam, the "5" is reproduced, and the "6" is 2–3 units (≈ 0.1 of peak) too bright; the best model is the beam plus a component of 7–12% of peak located 27–31 s earlier ($\ln\mathscr{L} = -5.4$ at 35 s, -9.6 at 36.5 s), while a smooth fade (-29 to -50) and a resolved source (-23.6) fail. Neither horn's measured response reproduces that excess: used directly as templates both predict 3.5 ± 1.5 for the "6" record ($\ln\mathscr{L} = -52$ to -54), so a 10% leading shoulder is absent on both horns at the $\approx 2\sigma$ level of this scan (noise ≈ 0.05 of peak). Ehman reports that the 1994 OY 372 scans showed "a plateau only about 10 dB down" on each side of both horns with unequal amplitudes on opposite sides, so an asymmetric coma shoulder remains possible instrumentally; it is not seen at the required level in 1977. Finally, the measured separation moves the first-beam position to RA $19^{\mathrm{h}}28^{\mathrm{m}}03^{\mathrm{s}} \pm 5^{\mathrm{s}}$. The physical reading becomes: an unresolved source that was $\approx 2\times$ brighter during the first ten seconds of the crossing (or had a 10% companion 6–7′ east) and switched off, or on, within 181 s on either side; the horn probability of Sect. 3.1, which rests on the duration prior rather than on the fade, is unchanged.

3.4 The frequency reference

Méndez et al. (2025) obtain the centre-channel sky frequency as $f_c = f_{2\mathrm{LO}} - R + 1420.4056$ MHz with the 2nd-LO reading $f_{2\mathrm{LO}} = 120.185$ MHz at the peak and a reference $R = 120.1$ MHz (before 1977 December 13), and, with the inverted channel axis, channel 2 at $f_c + 0.235$ MHz $= 1420.726$ MHz. Our astropy reconstruction of their velocity chain is exact: 1420.726 MHz topocentric is $-67.6\,\mathrm{km\,s^{-1}}$; the barycentric correction is $-16.9\,\mathrm{km\,s^{-1}}$ and the standard solar motion $+9.9\,\mathrm{km\,s^{-1}}$, giving -84.5 and $-74.5\,\mathrm{km\,s^{-1}}$ against their -84 ± 1 and -74 ± 2.

The LO reading, however, is itself a check on R. Méndez et al. (2026) state that the 2nd LO was software-tuned to hold the Galactic standard of rest. For the Wow! direction and epoch the frame offsets of the centre channel are -0.080 MHz (barycentric), -0.033 MHz (LSR) and $+0.166$ to $+0.180$ MHz (GSR with $\Theta_0 = 220$–$236\,\mathrm{km\,s^{-1}}$). Subtracting the GSR offset from the reading gives $R = 120.005$–120.019 MHz, i.e. the design value of 120.0 MHz; the LSR gives 120.218, the barycentre 120.265, and only a barycentric correction applied with the wrong sign returns 120.105 MHz. Either the LO tracked the GSR as documented and $R = 120.0$ MHz, or $R = 120.1$ MHz and the LO was not tracking the GSR at that moment; the two 2025–2026 papers cannot both be right about this. The three candidate references give channel-2 frequencies of 1420.726, 1420.826 and 1420.926 MHz and $V_{\mathrm{LSR}} = -74$, -95 and $-116\,\mathrm{km\,s^{-1}}$. Méndez et al. (2025) fixed the post-December-13 reference (119.9 MHz) by matching the Wow2 and Wow3 clouds of January 1978 to HI4PI, which is sound, but those clouds post-date the software change and so never calibrated the August 1977 value; for that value the paper is internally inconsistent, giving 120.1 MHz "(not 120.0 MHz)" in one passage and describing the December 13 update as moving the LO "from 120.0 MHz to its correct value of 119.9 MHz" in another. The LO reading at the peak, under the documented GSR tracking, agrees with the second statement. A plot of LO reading minus predicted frame offset against date over 1977–78 would settle both the frame and R to $\pm 2\,\mathrm{km\,s^{-1}}$.

Two further points. At $l = 11.6°$, $b = -18.8°$ a flat rotation curve ($R_0 = 8.15$ kpc, $\Theta_0 = 236\,\mathrm{km\,s^{-1}}$; Reid et al., 2019) gives $V_{\mathrm{LSR}} = +6\,\mathrm{km\,s^{-1}}$ at 1 kpc, +23 at 3 kpc and +172 at 8 kpc, turning negative only beyond 17 kpc where the sightline is 5.5 kpc below the plane: all three candidate velocities are forbidden for circular motion, and any H I host is intermediate- or high-velocity gas. The field lies inside the footprint of the HVC complex Galactic Center Negative ($l \simeq 0$–$70°$, $b \simeq$

$-60°$ to $+10°$, $V_{\rm LSR}$ from -350 to $\simeq -70\,{\rm km\,s^{-1}}$; Winkel et al., 2011), 243 compact and mostly unresolved clumps with almost no diffuse envelope, a natural host population at these velocities. Finally, the Galactic latitude printed for the positive horn in Table 4 of Méndez et al. (2025), $-17.85°$, does not follow from its RA and Dec, which give $b = -18.82°$ (the negative-horn value is reproduced); a $1°$ error propagates into any cross-match.

4 Origin simulations

Each scenario was tested against five facts: a 38 s point-like crossing; the skew; detection in one beam and $< 2\sigma$ in the other 172 s away; the line frequency; and no recurrence in 178 further Big Ear passes (1977–84) plus 193.5 h of targeted follow-up. Figure 3 shows the two-beam constraint and the repeater likelihood.

Steady sources. Any steady celestial source is excluded: the second beam would have printed ≥ 30 and 178 later passes stayed blank. A steady source seen through diffractive interstellar scintillation (gamma-distributed intensity, independent scintles between beams and days) has a maximum likelihood of 3×10^{-12} at a mean flux of 11 Jy for modulation index unity, and 10^{-44} once the follow-up sensitivity caps the mean flux. Extended H I clouds of the Wow2/Wow3 type (Méndez et al., 2025) give 50–60 s crossings at $\sim 3.5\sigma$ in both beams every day.

Repeaters. With the exact one-beam-only geometry (a pulse covers one beam and misses the other only if it starts or ends within a 104 s window), 90 Big Ear visits give a maximum likelihood of 27% for a stochastic repeater, 1.0% once the 193.5 h of follow-up are added (best fit $D \approx 170$ s, rate $\approx 0.12\,{\rm d^{-1}}$; 2.6σ tension), reproducing Kipping & Gray (2022) (32%, 1.8%). A further ≈ 750 h would push every repeater below 0.27%. One caveat they did not raise: META's 400 kHz band centred on the rest frequency would not have contained 1420.726 MHz, so 8 of the 193.5 h may not count; Hobart's 2.5 MHz band (Gray & Ellingsen, 2002), the ATA and the GBT did cover it. Strictly periodic repeaters are excluded below 14–40 h (Gray & Ellingsen, 2002; Harp et al., 2020).

Moving objects and interference. The width constrains the source to the sidereal rate within 4%, i.e. $|\mu| < 0.6''\,{\rm s^{-1}}$: this excludes Earth satellites and aircraft but allows $435\,{\rm km\,s^{-1}}$ transverse at 1 AU and anything slower than $14°\,{\rm d^{-1}}$ among solar-system bodies, which the two-beam test then removes unless the object moved $> 7''\,{\rm s^{-1}}$ in declination (a circular orbit radius of $\approx 70\,000$ km). For terrestrial interference, the often-quoted $(1/32)^6 \approx 10^{-9}$ is the probability of one exact character string, not of a beam-shaped pattern; a Monte Carlo of 400 lognormally modulated smooth bursts per correlation time (15, 40, 100 s) passed through the recorder finds that 0.3–1.5% mimic a crossing in shape and width. Against that stands the base rate: the Ohio archive holds $> 40\,000$ events above 3σ that passed the antenna-pattern filter, and a 30σ event in the "southern hot spot" within a year of Wow! (Méndez et al., 2026).

5 A natural one-off transient

Only one natural mechanism satisfies all five facts. Every broadband emitter (pulsar giant pulses, magnetar radio bursts, flare continua) is excluded by the blank adjacent channels; electron-cyclotron bursts from flare stars would need the star within 0.3 pc; recombination and molecular lines are ≥ 4 MHz away; extragalactic H I is redshifted. What remains is stimulated or coherent 21-cm emission from cold hydrogen, pumped on a timescale of a minute.

Maser or coherent burst. The gain arithmetic does not exclude an ordinary maser; the fade does. With $\tau = N_{\rm H}/(1.82 \times 10^{18}\,T_s\,\Delta v)$, a cloud with $N_{\rm H} = 2 \times 10^{20}\,{\rm cm^{-2}}$, $\Delta v = 2\,{\rm km\,s^{-1}}$ and a 1% hyperfine inversion ($|T_s| \approx 1.7$ K) has $|\tau| \approx 8$ and amplifies the 6 K sky to $\approx 18\,000$ K, the brightness temperature a $1.9'$ source needs for 256 Jy. The widely quoted requirement of $10^3\times$ normal densities (Méndez et al., 2024, eq. 15) assumes an inversion of 6×10^{-7}, the limit for a blackbody-shaped pump; a steep Lyman-α profile allows 10^{-3}–10^{-2} before multiple scattering thermalises it. But an amplifier cannot switch off faster than its light-crossing time: a 17 s fade bounds the path to 5×10^{11} cm (0.03 AU) and the density to $4 \times 10^8\,{\rm cm^{-3}}$—unless the pump sweeps through the gas toward the observer, so that pump and 21-cm photons travel together and the whole path contributes within the pump's duration. That gain-swept configuration is the one in which the published 21-cm superradiance model (Rajabi & Houde, 2016) has a "minutes" regime: $T_R = 1$ s, delay $\langle \tau_D \rangle = 66$ s, bursts of a few hundred seconds from samples $\sim 10^{14}$ cm long, $n_{\rm H} = 10\,{\rm cm^{-3}}$, 1% inversion—numbers that match the Wow! fade, duration and two-beam window without tuning, with isotropic energies of only 3×10^{29} erg at 1 kpc.

Two unproven requirements. Collective emission needs velocity coherence $\Delta v \lesssim \lambda/(2\pi T_R) \approx 3\,{\rm cm\,s^{-1}}$ inside each coherent sample, four to five orders of magnitude below the $350\,{\rm m\,s^{-1}}$ thermal spread at 15 K; Rajabi & Houde (2016) acknowledge this and do not resolve it. And a coherent sample of width w radiates into a fraction $\approx 3 \times 10^{-14}$ of the sky for $w = 300$ km, so the emission direction must be fixed by the pump (radially away from the trigger) and Earth must lie on the trigger–knot line within the knot's radius: probability $(s/r)^2/4$, i.e. 2% for a 15 AU knot 50 AU from its star, 10^{-9} at 1 pc, 10^{-12} at 50 pc.

Pump and trigger. Wouthuysen–Field pumping (Wouthuysen, 1952; Field, 1958) fast enough to set up the inversion within ~ 60 s requires a Lyman-α flux of $\approx 17\,{\rm erg\,cm^{-2}\,s^{-1}}$ at the knot: within 46 AU of a 10^{35} erg M-dwarf superflare,

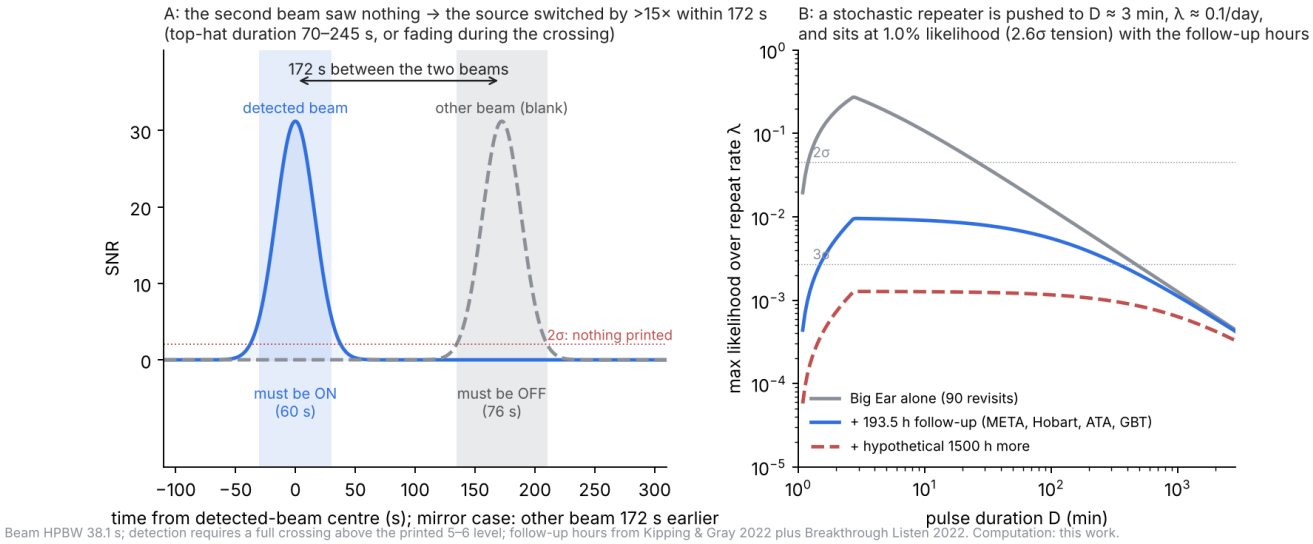


Figure 3: (A) The two-beam constraint: a source detected at $> 30\sigma$ in one beam must have been below 2σ when the other beam crossed it 172 s earlier or later, so a top-hat emitter lasted 70–245 s or faded by $> 15\times$ within three minutes. (B) Maximum likelihood over repeat rate for a stochastic repeater of pulse duration D, with exact single-pulse geometry, 90 Big Ear visits and the follow-up hours of Kipping & Gray (2022) plus Breakthrough Listen (2022).

150 AU of a 10^{37} erg giant flare, 0.13 pc of a magnetar giant-flare tail (at 10^{-4} Lyman-α efficiency) or 50 pc of its 0.2 s spike. Big Ear's exposure (14 sky-minutes of $8' \times 40'$ beams over 1977–84, more than all modern H I surveys combined) and one event imply an all-sky rate of ~ 100 events per day at ≥ 250 Jy (90% range 5–300). Stellar superflares occur at $\sim 3 \times 10^5$ d^{-1} Galaxy-wide, so the rate is met if $\sim$0.1–2% of superflaring stars host a cold H I knot within ~ 50 AU; the magnetar trigger proposed in the literature fails on pump efficiency (a 10^{18} cm^{-2} knot is transparent to keV X-rays), on duration (the spike lasts 0.2 s against a 60 s pump requirement) and on alignment, together by more than ten orders of magnitude. The natural hypothesis therefore predicts a dense, cold, tens-of-AU knot of the tiny-scale-atomic-structure type (Heiles, 1997, 2007) at the position of a flaring star whose radial velocity is -84, -105 or -127 km s^{-1}, a flare that reached Earth within a minute of 03:16 UTC on 1977 August 16, and a 21-cm light curve that copies the flare's. The accessible Helios-2 burst papers of 1979 list five triangulated events in 1976–77 (none in August 1977); the complete Vela 5/6 record for that night remains to be checked. It also predicts, for any smooth luminosity function, hundreds to 10^5 one-pass narrow spikes at $|V_{\rm LSR}| > 50$ km s^{-1} in the RFI-rejection products of ALFALFA, FAST and HI4PI.

6 An artificial source

The received power flux, $\geq 2.6 \times 10^{-20}$ W m^{-2} regardless of the signal's intrinsic width, requires EIRP $\geq 3 \times 10^{14}\,(d/\mathrm{pc})^2$ W: 2.4 GW into an Arecibo-class dish at 10 pc, 240 GW at 100 pc, 7 TW at 548 pc; a 3 km phased array needs 100 times less (Table 2). Energy is not a constraint for a civilization a century or two ahead of ours. Nor would our own technology qualify: Earth's planetary radars and the 1974 Arecibo message would deliver the Wow! flux only to 0.26–0.6 pc, all of Earth's electricity radiated isotropically to 0.08 pc, and a Kardashev-I civilization to 24 pc; the one Earth analogue reaching stellar distances is a Starshot-class propulsion beamer ($\sim 10^{20}$ W EIRP for minutes per launch), visible to ≈ 570 pc. Leakage is thus excluded, and the emission was collimated toward us.

Table 2: Transmitter power for the Wow! flux. EIRP $= 3 \times 10^{14}(d/\mathrm{pc})^2$ W.

d	EIRP (W)	300 m dish	3 km array	30 km array
100 AU	7×10^7	6 W	–	–
1 pc	3×10^{14}	24 MW	240 kW	2.4 kW
10 pc	3×10^{16}	2.4 GW	24 MW	240 kW
100 pc	3×10^{18}	240 GW	2.4 GW	24 MW
548 pc	9×10^{19}	7 TW	71 GW	710 MW
1 kpc	3×10^{20}	24 TW	240 GW	2.4 GW

The difficulty is timing. Big Ear watched two $8' \times 40'$ patches and spent ~ 3 d on each of ~ 300 declination strips, so a transmission of duration D at a random moment was caught with probability $\approx 2(D+72\,\mathrm{s})/(300 \times 86\,164\,\mathrm{s})$: 2×10^{-5} for a 3 min ping, 3×10^{-4} for an hour. A sweeping beacon of the cost-optimised type (Benford et al., 2010), dwelling 2–4 min on each of 10^4–10^5 targets, is the only artificial variant with a quantifiable likelihood, and it peaks at 1–2% (revisit of order a week) once the null follow-ups are counted. Accidental high-gain beams (propulsion or radar) and in-system probes need no repetition but cannot be re-observed. The profile's

gradual 15–20 s decline fits a beam sweeping off us or a power ramp-down ($\ln \mathscr{L} = -4.7$) better than a switched transmitter (-9.8).

A transmitter radiating at the H I rest frequency in its own frame fixes its host's heliocentric velocity at -84.5, -105.6 or $-126.7\,\mathrm{km\,s^{-1}}$ for the three references; a signal placed at the line in the solar-barycentric, LSR or GSR frames would have appeared at 1420.326, 1420.373 or 1420.571 MHz, none of which was observed. A Gaia DR3 census of the first-beam box ($3' \times 40'$ around $19^{\mathrm{h}}27^{\mathrm{m}}55^{\mathrm{s}}$, through the Harvard VizieR mirror) returns 33 stars with radial velocities, spanning -200 to $+120\,\mathrm{km\,s^{-1}}$ with a dispersion of $69\,\mathrm{km\,s^{-1}}$, so that 0.3–0.5 chance matches per candidate velocity were expected at $\pm 3\,\mathrm{km\,s^{-1}}$; none occurred. The nearest approaches are a $G = 12.9$ star at 870 pc with $-75.8\,\mathrm{km\,s^{-1}}$ and two distant halo stars at -122.5 and $-130.7\,\mathrm{km\,s^{-1}}$; nothing lies near $-105.6\,\mathrm{km\,s^{-1}}$. The second-beam box ($19^{\mathrm{h}}25^{\mathrm{m}}02^{\mathrm{s}}$) holds 40 such stars and one match: a $G = 14.3$, bulge-distance giant (parallax 0.12 mas) $16'$ north of the box centre with $-83.98\,\mathrm{km\,s^{-1}}$, $0.5\,\mathrm{km\,s^{-1}}$ from the -84.5 prediction; in a field with an $81\,\mathrm{km\,s^{-1}}$ velocity spread 0.57 chance matches were expected at that velocity, so this is a target for follow-up, not evidence. Both censuses constrain only $G \lesssim 14$, i.e. giants and F–K dwarfs to $\sim$1 kpc; M dwarfs beyond $\sim$100 pc have no Gaia velocities. GASS and LAB spectra at both positions (0.3° beam) show a single local H I component at $V_{\mathrm{LSR}} \approx 0\,\mathrm{km\,s^{-1}}$ and nothing above ≈ 0.2 K at -74, -95 or $-116\,\mathrm{km\,s^{-1}}$: no extended gas and no GCN clump lies in the beams, as the circumstellar-knot variant of Sect. 5 predicts and the extended-cloud variant does not. The Sun-like star 2MASS J19281982−2640123 proposed by Caballero (2022) and observed by Breakthrough Listen (2022) lies $5.5'$ east and $17'$ north of the revised first-beam position (beam response ≈ 0.12, requiring ≥ 2000 Jy) and has a Gaia radial velocity of $-13.7 \pm 6\,\mathrm{km\,s^{-1}}$; it is no longer a strong candidate. The expected nearest star in the two $1.5' \times 40'$ error boxes is at ≈ 140 pc, where the signal left in 1521, Earth's 2012 Arecibo reply arrives in 2470 and an answer could not arrive before 2926. At any distance beyond ≈ 7 pc the sender could not have known of human technology.

7 Discussion: a probability assessment

We group the hypotheses as H_{A} (anthropogenic or instrumental), H_{N} (natural one-off transient) and H_{E} (artificial extraterrestrial), and argue for posterior odds of roughly 50–60%, 25–35% and $\lesssim 10\%$. The argument is a product of factors, each of which we state so that readers can substitute their own.

Base rates. The same instrument logged $> 40\,000$ narrowband events passing its pattern filter, with a 30σ event elsewhere within a year, and the recorder is known to interact with gain changes (Méndez et al., 2026); every other candidate technosignature in six decades has resolved to interference. The prior odds for H_{A} against the union of the other two are therefore large, of order 10:1 or more, before the Wow!-specific evidence is considered. For H_{N} the prior is the probability that a mechanism never observed in 70 years of 21-cm astronomy exists at the required rate, which we put at 0.1–0.3; for H_{E} the prior is the probability that a transmitting civilization lies within a few hundred parsecs along this sightline, which no current data constrain to better than 10^{-3}–10^{-1}.

Shape and width. A beam-matched width (4%) and a 38 s crossing are automatic for any celestial point source and occur for $\sim$1% of random smooth bursts. This is a likelihood ratio of $\sim$100 for celestial over terrestrial, per burst; multiplied by dozens of strong bursts in the archive it leaves H_{A} with an expected $\sim$0.1–0.5 Wow!-shaped impostors over the program, i.e. not improbable.

The fade and the blank second beam. These favour any intermittent source over any steady one, which is why all steady sources are excluded (Sect. 4), but they do not discriminate between H_{A}, H_{N} and H_{E}: interference is intermittent by nature, a pumped burst ends, and a beam sweeps on. The gradual decline mildly favours a physical decay or a beam sweep over a switched transmitter.

Frequency. Being within 0.3 MHz of the hydrogen line is predicted by H_{N} (the line is the only natural candidate) and is a plausible deliberate choice under H_{E}, but it carries no weight against H_{A}: any interference Big Ear could see had to fall in its 500 kHz window, and the base rate above already counts in-window events. The velocity, forbidden for disk gas, costs H_{N} a factor of a few (a halo-velocity host is rarer than a disk one) and costs H_{E} nothing.

Non-recurrence. 178 Big Ear passes and 193.5 h of follow-up are neutral for one-off versions of all three hypotheses and penalise repeating ones; the only quantifiable repeater, a sweeping beacon, retains 1–2%.

The timing coincidence. This is where H_{E} loses. Every artificial scenario must explain why two small beams happened to be on the source during the minutes it pointed at us: 10^{-5}–10^{-3} for one-off emission, $\sim 10^{-2}$ for the best repeater. H_{N} pays the same coincidence but has it priced into the rate (one event in 14 sky-minutes), which is a statement about the all-sky rate rather than about luck; H_{A} pays nothing, because interference is local and frequent.

The physics penalty. H_{N} in turn pays for two unestablished requirements ($\mathrm{cm\,s^{-1}}$ coherence; an aligned circumstellar geometry) and for a host population—dense neutral knots within tens of AU of flaring stars—that has never been observed; we assess this at a factor of 3–10 against it. H_{E} pays no physics penalty; H_{A} pays the penalty of having no identified emitter: a frequency-stable 10 kHz source in a protected band, lasting

minutes, strong enough to look like 250 Jy, never heard again, which we assess at a factor of 3–10 as well.

Combining: H_{A} starts far ahead on base rate and loses a factor of order 10 on shape and 3–10 on the missing emitter; H_{N} starts behind, gains on shape and frequency, and loses 3–10 on physics; H_{E} starts behind, gains on shape and frequency, and loses 10^2–10^4 on timing. The outcome is the ordering above, with H_{A} and H_{N} within a factor of two of each other and H_{E} an order of magnitude below. We regard the split between H_{A} and H_{N} as genuinely open and the position of H_{E} as robust to any plausible choice of priors short of assuming a nearby civilization.

Limitations. The per-record noise is taken as $1/\sqrt{10}$ of the 1 s noise; if the 1 s samples were correlated the asymmetry significance drops (the skewed model's advantage falls from 21 to 3 log-units for a threefold larger noise) but the preference remains. The N50CH freeze threshold is undocumented and was scanned over 3–15σ. The LO argument rests on the staff's statement of GSR tracking and on the published formula; the LO logs would resolve it. No H I spectrum at the position and no Gaia census could be retrieved in this work.

8 Tests

Ranked by cost: (1) per-horn beam widths and leading/trailing half-widths from the 1977 calibrator transits (done here for OV-221 from one scanned chart; a higher-resolution scan and the other calibrators would sharpen the $\approx 2\sigma$ shoulder limit); (2) LO reading minus predicted frame offset against date for 1977–78 (days; settles R and the velocity); (3) a Gaia DR3 census of the two error boxes with radial velocities, flagging -84.5, -105.6 and -126.7 ± 3 km s^{-1} and flare activity (done here for both boxes: one chance-level match in the second box; the first box should be repeated $2'$ east and the faint M-dwarf population remains); (4) VLA/MeerKAT H I absorption at both positions at the three velocities (single-dish spectra are null); (5) the Vela 5B/6A record for 03:13–03:16 UTC 1977 August 16 (the published Helios-2 events exclude a triangulated burst that night); (6) a one-pass narrow-spike search in survey RFI products; (7) ≥ 750 h of continuous, Hz-resolution, full-Stokes monitoring of both $40'$ boxes, for which the flux makes a small-dish network sufficient.

9 Conclusions

Six quantized numbers still contained one unused observable—their asymmetry—and the beam measured on the night sharpens what it means: the core is the 36-s beam, unresolved, and the first record carries a $\approx 10\%$ excess that neither horn's measured response produces, so the source was briefly brighter (or accompanied) at the start of the crossing and switched within 181 s on either side; the detection sits in the first beam, now at RA $19^{\mathrm{h}}28^{\mathrm{m}}03^{\mathrm{s}}$, with probability 0.89. The frequency rests on a 2nd-LO reference that the LO reading itself contradicts under the documented tracking mode, and the size limit rests on an unmeasured beam width; both are resolvable from existing records. No origin is identified. The natural and anthropogenic readings are within a factor of two of each other; the artificial reading is not excluded by any physics but is an order of magnitude behind on the single question of why we were listening.

Data and code availability

All fits, simulations and geometry calculations are reproduced by seven short Python scripts (numpy, scipy, astropy 8) released with this manuscript; every number in the tables regenerates in under 30 minutes.

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